

REQUIREMENT ANALYSIS

Metro Ticket Generating System in ServiceNow

Customer Journey • Requirements • User Stories • Technology Stack

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Roll Number	238W1A4259
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	2 Marks

This phase turns the outcomes of ideation into clear customer needs, functional and non-functional requirements, user stories, data movement, and the technology stack behind the ServiceNow solution. The scope covers the Service Catalog booking flow, fare logic, payment options, and the QR ticket widget; there is no separate ticket database, no Flow Designer automation, and no Requested Item record kept for a booking.

1. Customer Journey Map

The journey below maps out what a metro passenger goes through when using the proposed digital booking service.

Journey Stage	Customer Action	Touchpoint	Pain Point	Expected Experience	System Opportunity
Need to travel	Decides where the trip starts and ends	Metro service	Wants quick access to a ticket	A booking flow that is clear and easy	Service Catalog entry
Start booking	Opens Book A Metro Ticket	Service Portal	Manual queues can take a while	A simple digital form	Catalog item
Enter journey details	Picks starting/going-to stations, journey type, and passenger count	Booking form	Fare amounts can feel unclear	Fare works itself out automatically	Client-side fare logic
Payment	Picks UPI, Card, or Others	Payment section	Preferred payment method can differ	The right payment field shows up	Catalog UI Policy
Confirm	Reviews the order	Order confirmation	Wants to feel sure before submitting	A clear confirmation screen	Catalog workflow
Receive ticket	Views the completed booking and a QR code on screen	Confirmation / QR display	Needs a ticket that's easy to access	A digital QR ticket shown right away	QR widget

2. Solution Requirements (Functional & Non-functional)

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / SubTask)
FR-1	Metro Ticket Booking	Offer a Service Catalog item called Book A Metro Ticket.
FR-2	Station Selection	Let passengers pick Starting From and Going To stations using the Metro Station's Details reference data.
FR-3	Journey Selection	Support both Single journey and Return journey options.
FR-4	Passenger Selection	Allow between 1 and 4 passengers to be chosen.
FR-5	Fare Calculation	Work out the fare automatically from the source, destination, journey type, and passenger count.
FR-6	Payment Support	Offer UPI, Card, and Others as payment modes.
FR-7	Conditional Payment Details	Show the Enter Payment Mode field only when Others is chosen.
FR-8	QR Ticket	Produce a QR representation of the ticket on submission.
FR-9	Confirmation Display	Show the journey, fare, payment, and QR details on the confirmation screen for the passenger to review; no separate record is stored afterward.
FR-10	Validation	Stop the same station being picked for both source and destination.
FR-11	Client-Side Automation	Use Catalog Client Scripts to run fare calculation and QR generation without a server-side flow or database.

Non-functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The booking form should stay simple and easy for passengers to follow.
NFR-2	Security	Rely on ServiceNow roles/ACLs and platform access controls to protect custom data.
NFR-3	Reliability	Fare working-out and QR generation should behave consistently for valid inputs.
NFR-4	Performance	Form actions and QR generation should finish without needless delay.
NFR-5	Availability	The solution should lean on the availability built into the ServiceNow platform.
NFR-6	Scalability	Master station data should be able to grow to include more stations and fare rules over time.

3. Data Flow Diagram & User Stories

DFD — Level 1 textual representation

Entity / Process	Data Flow	Data Store / Output
Passenger	Provides station choices, journey type, passenger count, and payment choice	Feeds the Book A Metro Ticket catalog form
Catalog Form	Passes booking details into the fare logic	Populates the fare fields
Fare Logic	Combines source, destination, journey type, and passenger count	Produces the single/return amount
Payment UI Policy	Reacts when Mode of Payment = Others	Reveals the Enter Payment Mode field
QR Generation	Turns a ticket identifier/URL into a QR image	Feeds the QR widget / digital ticket
Confirmation Screen	Displays the completed journey, fare, payment, and QR details	Shown to the passenger; no record is kept beyond the session

User Stories

User Type	Epic	USN #	User Story / Task	Acceptance Criteria	Priority	Release
Passenger	Ticket Booking	USN -1	As a passenger, I want to open the metro ticket catalog item and fill in my journey details.	The booking form opens correctly and accepts valid entries.	High	Sprint-1
Passenger	Station Selection	USN -2	As a passenger, I want to choose my starting and destination stations.	Both station fields offer valid station records.	High	Sprint-1
Passenger	Journey Type	USN -3	As a passenger, I want to pick either a single or return journey.	The chosen journey type is saved and used to work out the fare.	High	Sprint-1
Passenger	Passenger Count	USN -4	As a passenger, I want to specify how many people are travelling.	Values from 1 to 4 can be selected.	High	Sprint-1
Passenger	Fare Calculation	USN -5	As a passenger, I want to see the fare worked out automatically from my choices.	The correct single/return amount is filled in.	High	Sprint-1
Passenger	Payment	USN -6	As a passenger, I want to choose between UPI, Card, or Others for payment.	The chosen option is saved, and Others reveals its extra field.	High	Sprint-2
Passenger	QR Ticket	USN -7	As a passenger, I want to get a QR version of my ticket on the confirmation screen.	A QR code is shown as part of the ticket flow.	High	Sprint-2
Admin	Access Control	USN -8	As an administrator, I want to manage who can access metro station data.	Table ACLs/roles protect the custom table as intended.	High	Sprint-2

4. Technology Stack (Architecture & Stack)

The technical architecture centres on ServiceNow: the Service Catalog handles the passenger-facing request, Catalog Client Scripts drive fare and validation logic, a custom table holds the station master data, and a QR service/widget presents the digital ticket. The booking flow finishes with the QR ticket shown on the confirmation screen; there is no separate ticket database, Flow Designer automation, or Requested Item record kept for the booking.

S.No.	Component	Description	Technology
1	User Interface	The passenger's booking form and confirmation screen	ServiceNow Service Portal / Service Catalog
2	Application Logic-1	Fare working-out and same-station validation	Catalog Client Script (onChange) / JavaScript
3	Application Logic-2	Showing payment details conditionally	Catalog UI Policy
4	Application Logic-3	QR code generation on submission	Catalog Client Script (onSubmit) / QR Server API
5	Database	Station master data	ServiceNow custom table: u_metro_station_details
6	Digital Ticket UI	Displaying the QR code	ServiceNow Service Portal widget
7	Security	Controlling access	ServiceNow roles and default ACLs
8	Infrastructure	The application platform	ServiceNow cloud platform

Application Characteristics

S.No.	Characteristics	Description	Technology
1	Open-Source Frameworks	No extra open-source framework is needed for the ServiceNow build.	ServiceNow platform components
2	Security Implementations	The custom station table relies on the default ACLs ServiceNow creates automatically, with platform authentication and authorization also applying.	ServiceNow ACL / RBAC
3	Scalable Architecture	Station master data is kept separate from the catalog item itself, so more stations or fare rules can be added without changing the item.	ServiceNow tables + Service Catalog
4	Availability	The hosted ServiceNow platform supplies the application's runtime.	ServiceNow
5	Performance	Client-side fare working-out and QR generation avoid extra server calls during booking.	JavaScript + ServiceNow Service Catalog

Requirement Analysis Outcome

The requirements lay out a traceable path from passenger need, through catalog booking, fare working-out, and payment, to a QR ticket shown on the confirmation screen at the end of the booking, with no separate database, automation flow, or Requested Item record kept afterward.